

Fig 3: Basic compiler

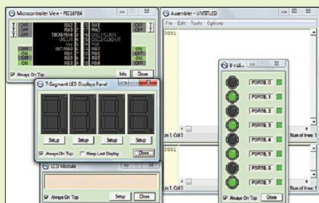


Fig 4: Modules and interfaces

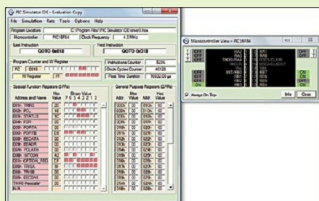


Fig 5: Timer

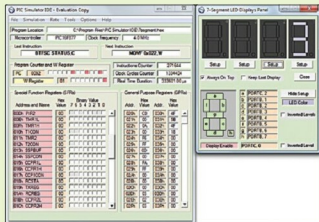


Fig 6: 7-segment LED

program is opened. From next time onwards, the file can be opened as normal by clicking the shortcut in Start menu or double-clicking the program file using left mouse button and the program gets launched.

The default storage location of program files is the PIC Simulation IDE subfolder in Program Files folder. Some example files to work with are stored in this subfolder. Modifying these files requires admin rights, without which the error message "Run-time error '75': Path/File access error" appears. It also contains three manuals for the users: for getting started, compiler reference and for external modules. Modifying files stored in Program Files folder requires admin rights.

To permanently set the program to run with administrator privileges, right-click on the program icon, select Properties, click on the button Advanced and tick the Run as administrator check box. Administrator rights are not required if the project files are stored in a user private folder (for example: My Documents.)

Getting started

The main window (refer Fig. 1) shows the location of the program being executed, name of the PIC selected, clock frequency, mnemonics of the instruction last executed and the one about to be executed, instructions and clock cycles counter, real-time duration of the simulation and the statuses of internal registers.

The File menu has options to load a program to the PIC memory, clear memory and save memory. From the Simulation menu, you can start or stop a simulation, execute the next step of simulation (when Step By Step simulation rate is selected), or run to the next Basic statement for programs generated by integrated Basic compiler. The user can set the simulation rate to Step By Step, Slow, Normal, Fast, Extremely Fast, and Ultimate from Rate option in the menu bar.

The Tools option (Fig. 2) gives a variety of modules and interfaces to choose from. Basic compiler opens the integrated Basic compiler editor window, as shown in Fig. 3. The Basic Compiler Reference Manual in Help menu of main window (or Options of Basic compiler editor window) contains more information on this module.

The Assembler command starts integrated assembler. In this window, the assembler source files can be edited and assembled. A similar Disassembler option starts disassembling automatically when this option is selected. Once the operation is complete, the output file is displayed and the user is prompted to save it.

Other main modules and interfaces include Microcontroller View, EEPROM Memory Editor, 8 x LED Board, 7-Segment LED Displays Panel, Stepper Motor Phase Simulator, Oscilloscope, Signal Generator, LCD Module, etc. Fig. 4 shows the screenshot of some of these modules.

Example 1: Timer0 module simulation

The easiest example file given in the package is that of an 8-bit Timer0 module of PIC16F84. The BASIC file timer0.bas uses Timer0 module interrupts to periodically change the value on Port B pins. The assembler source file timer0.asm was generated using Basic compiler integrated with the tool and the hexadecimal source file timer0.hex was generated using the integrated assembler:

```
TRISB = 0x00 'set all PORTB pins as outputs
PORTB = %11111111 'make all PORTB
```